Avantika Varma

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EDUCATION

University of California, Davis (UC Davis) Davis, CA
Bachelor of Science in Computer Engineering 2023 – 2027
Minor in Business

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, matplotlib, SciPy, PyTorch), C++, C, Verilog, RISC-V Assembly, MATLAB, HTML, Streamlit, Qiskit, PyVISA

EDA & CAD: Cadence, Synopsys, Ansys, Altium, OrCAD, LTspice, Quartus, ModelSim, Blender

Hardware & Lab: Microcontrollers (MSP430, Arduino), TI-CC3200, waveform generators, power supplies, Ethernet/socket communication, PCB testing, oscilloscopes, multimeters, CAEN digitizers

Tools & Systems: Git (GitLab), Linux/Unix, Vim, YAML, Excel, PowerPoint, Jira, VS code

EXPERIENCE

Solidigm Rancho Cordova, CA
NAND Design Automation Engineering Intern March 2026 – Present

- **Automation Scripting:** Developed Python and Bash scripts to automate regression flows for Verilog design testing in a Unix environment, reducing manual verification overhead and improving workflow efficiency.
- **Verification Infrastructure:** Configured and optimized Cadence EDA simulation workflows to accelerate bug identification and strengthen validation coverage for NAND flash design flows.
- **AI Workflow Automation:** Built automated data pipelines and machine learning scenarios in Dataiku to analyze large-scale simulation logs, improving anomaly detection and predictive failure analysis.
- **Language Modernization:** Migrated legacy Intel Perl automation scripts to Python, improving maintainability, and enabling integration with modern AI/ML tooling for downstream analysis.

University of California Davis College of Engineering Davis, CA
Peer Advisor March 2026 – Present

- **Academic Leadership:** Mentored undergraduate engineering students on degree requirements, course sequencing, and academic planning across demanding STEM curricula.
- **Cross-Functional Communication:** Served as a liaison between students and faculty, translating academic policies into clear, actionable guidance for peers.

California-Pacific-Northwest AI Hardware Hub Davis, CA
Research Intern, Integrated Nano Systems and Nano Devices Lab April 2025 – March 2026

- **Photonic Simulation:** Designed diatom-inspired photonic structures using Blender and CAD to optimize 3D geometries for light concentration and photon trapping in on-chip applications.
- **Grating Optimization:** Evaluated grating structures using FDTD simulations in Ansys Lumerical to characterize diffraction orders and electric field behavior.
- **Cleanroom Fabrication:** Trained in semiconductor fabrication processes and performed photolithography, spin coating, and e-beam metal deposition.

Tripathi Physics Lab (UC Davis) Davis, CA
Student Researcher September 2024 – June 2025

- **Digital Signal Processing:** Developed pulse shape discrimination (PSD) algorithms to differentiate neutron and gamma scintillation profiles based on timing characteristics.
- **Hardware Acquisition:** Integrated CAEN digitizers and SensL SiPMs for energy deposition capture and managed multi-format data acquisition across 16-channel signal splitters.

- **Hardware Automation:** Wrote Python automation scripts using PyVISA and socket programming to control waveform generators and power supplies via Ethernet.
- **Hardware Debugging:** Characterized custom amplifier systems and digitizer behavior using oscilloscopes, multimeters, and signal generators to debug hardware interfaces and assess signal integrity.

The Make Box (B2C STEM Startup) Davis, CA
Lead Marketing & Engineering Outreach Jan 2024 – Present

- **Strategic Pitching:** Secured $12,500 in funding and startup residency at Inventopia through the UC Davis Big Bang! Business Competition for Arduino-based robotics kits.
- **Mentorship:** Provided technical and strategic mentorship to early-stage student startups, supporting product validation and pitch development.

PROJECTS

Real-Time Stock Ticker Game April 2026 – June 2026
C++, Firmware, Microcontroller, I^2C, SPI, AWS IoT, APIs, LED Display

- Designed and programmed embedded firmware in **C++** for a microcontroller-based stock ticker that delivered real-time market-style updates to an LED display.
- Implemented I^2C and SPI communication protocols to interface reliably with components and drive display.
- Integrated external APIs and AWS IoT services to retrieve, transmit, and process live data streams, enabling responsive game behavior and cloud-connected functionality.

Die Counting Game on FPGA March – June 2025
Verilog HDL, Intel Quartus Prime, ModelSim, FPGA Hardware

- Designed and implemented a digital dice game on an Altera MAX FPGA using Verilog, with modular components for pseudo-random number generation and finite state machine (FSM) control logic for custom seven-segment displays.
- Applied verification methodologies in ModelSim by building test benches to validate logic correctness, timing behavior, and edge cases.
- Physical testing on FPGA hardware was performed to confirm signal integrity and verify that the synthesized logic was correctly mapped to the device resources.

AI-Powered Data Analytics Dashboard April 2025
Streamlit, Python, Gemini AI, Pandas, NumPy

- Developed a modular business intelligence platform in Streamlit and Python to transform raw nonprofit datasets into actionable insights through automated visualization.
- Integrated Gemini AI via API to support natural language queries, enabling users to generate summaries, visuals, and insights directly from their data.

Music-Following Robot January – March 2025
TI MSP430, C++, Code Composer Studio, Analog Electronics

- Designed an autonomous robot using an MSP430 microcontroller and implemented C++ firmware for real-time noise sampling and pulse width modulation (PWM) motor control.
- Analog Signal Conditioning: Built active filtering stages with operational amplifiers to isolate target musical frequencies and guide motion.
- Diagnosed and resolved circuit issues by testing continuity, voltage levels, and current flow across electronic components.

AWARDS & HONORS

- **Big Bang! Business Competition Winner** (2025) – $12,500 in funding and startup residency awarded
- **Best Project Award** (2025) – California-Pacific-Northwest AI Hardware Hub and CITRIS showcase
- **Dean's Honors List** (Fall 2023) – Awarded to the top 16% of the college